

Through the Lens of Privacy: Exploring Privacy Protection in Vision-Language Model Interactions on Smart Glasses

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Abstract

The integration of Vision-Language Models (VLMs) into smart glasses enables powerful interaction capabilities but also introduces significant privacy concerns. Through analysis of 376 real-life VLM interactions on smart glasses, we found frequent unintentional disclosure of sensitive information in the background, including people's faces, location cues, and screen content. To address this, we introduce a hand-gesture-based auto-cropping mechanism that allows users to control visual input shared with VLM, effectively filtering out sensitive data. The exploration study results show that our approach reduces explicit privacy leakage by 70.6% and inferential disclosure by 47.6% while maintaining VLM response quality. A preliminary user study (N=7) demonstrated that our method is intuitive and effective at protecting user privacy. This work offers a starting point for designing privacy-preserving interactions for VLM-enabled smart glasses.

CCS Concepts

• **Security and privacy** → **Privacy protections**; • **Human-centered computing** → Empirical studies in interaction design.

Keywords

Smart glasses, Vision-language model, Privacy, Wearer's perspective

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1 Introduction

Vision-Language Models (VLMs) have significantly expanded the capabilities of Large Language Models (LLMs) by enabling the processing of both visual and textual inputs [1, 23]. While VLMs are commonly accessed through smartphones and computers, their integration into smart glasses offers a more natural, hands-free interaction paradigm. For instance, users of Meta Ray-Ban smart glasses can simply say¹ "Hey Meta, (look and)"² tell me about this building" to receive instant responses from the integrated VLM.

While smart glasses offer convenience for interacting with VLMs, the always-on, always-seeing nature raises significant privacy concerns [7, 13, 15, 27], particularly as their form factor becomes increasingly indistinguishable from regular glasses [4]. Previous research has addressed privacy issues related to smart glasses' life-logging features [17, 34], the integration of VLMs introduces a new challenge: how to protect user privacy when interactions require transmitting potentially sensitive visual information to remote servers for processing. If left unaddressed, these privacy concerns could lead to another mass rejection, similar to the case of Google Glass [22].

To find a potential solution, it is crucial to understand how users interact with VLMs on smart glasses and identify patterns of sensitive disclosure. While previous research has investigated privacy leakage in LLMs from user perspectives [21, 37], understanding of VLMs is limited, particularly in the context of smart glasses, primarily due to the lack of real-life interaction datasets. Our research addresses this gap by answering the research question (RQ):

RQ1: What types of sensitive information are disclosed when users interact with VLMs on smart glasses, and what are the patterns of the disclosure?

We analyzed 376 real-life VLM interactions on smart glasses, collected through both public posts on the Internet and a structured questionnaire. Our analysis revealed three primary query types: (1) general scene understanding, (2) object/creature-specific queries, and (3) advice seeking. We also found that sensitive data, including

¹Meta AI on Meta smart glasses: <https://about.fb.com/news/2024/04/new-ray-ban-meta-smart-glasses-styles-and-meta-ai-updates/>

²New AI features on Meta smart glasses: <https://about.fb.com/news/2024/09/ray-ban-meta-glasses-new-ai-features-and-partner-integrations/>

facial information, location-revealing details, and screen content was frequently captured unintentionally in image backgrounds. Further investigation using gpt-4o³ confirmed that subtle visual cues in the background enabled accurate location inference in 65.6% of cases (21 of 32 had the correct inference within top 3 answers). These findings motivated our second research question:

RQ2: How can we design user-centric privacy protection mechanisms for VLM-enabled smart glasses that effectively mitigate the risk of sensitive information disclosure while preserving the utility?

Using the dataset with research permission, we explored whether cropping images to include only query-relevant regions could reduce privacy risks. The results showed cropping can reduce 70.6% of explicit sensitive information disclosure (such as bystander faces) and 47.6% of location inferability while maintaining the quality of VLM responses. Based on these findings, we proposed and implemented a hand-gesture-based auto-cropping mechanism that enables users to define precise visual boundaries for VLM processing. A preliminary user study (N=7) collected user feedback for future improvements..

In summary, this work makes the following contributions:

- A dataset of 376 real-world user conversations with VLMs on smart glasses, which we believe can benefit future VLM-related privacy research and interaction design.
- A promising user-centric mechanism that uses hand to control visual information shared with VLMs, with initial evaluation showing potential effectiveness.

2 Background and Related Work

The integration of VLMs into smart glasses represents a significant evolution in wearable computing⁴, building upon earlier developments in both ‘Data Glasses’ for capture and recording, and Augmented Reality (AR) Glasses for spatial computing. However, these devices’ always-on camera capabilities have raised privacy concerns for bystanders [13–15, 27]. Prior research has identified several key issues: users often underestimate potential privacy risks [7], devices lack clear recording indicators [16], and bystanders have limited ability to revoke consent after being captured [8]. These concerns can also create psychological burdens for wearers and potentially hinder widespread adoption [4].

Previous research on smart glasses privacy has proposed both design guidelines [12] and technical solutions, such as automatically disabling recording functions when detecting screens [17] or blocking captured faces [34]. However, these approaches only addressed specific scenarios and did not consider the privacy challenges introduced by VLM integration. Privacy challenges related to LLMs and VLMs can be classified into two types: memorization and inference. **Memorization risks** refer to language models, trained on massive datasets, retaining and revealing sensitive information from their training data [6, 36]. **Inference risks** refer to models’ ability to infer sensitive attributes such as location, gender, or income level from inputs, even if they lack obvious identifying information [29, 30].

Existing solutions for mitigating privacy risks fall into two categories: **model-side** and **user-side** protections. **Model-side protections** primarily focus on interventions during or after model training, such as data scrubbing [11], differential privacy (DP) algorithms [33], and applying safety filters [31, 35]. However, comprehensive removal of sensitive data is challenging and can negatively impact model performance [20]. Safety filters, on the other hand, can often be circumvented through various “jailbreak” techniques [2, 3]. **User-side protection** aims to prevent sensitive information from reaching the model in the first place. For example, Rescriber [40] automatically replaces sensitive data with placeholders, though this may weaken model understanding and does not address the risk of inference. PAPILLON [28] optimizes input prompts to reduce privacy leakage but relies heavily on prompt optimization model performance. While these solutions primarily address text-based interactions with LLMs, our work specifically targets the privacy challenges in VLM interactions on smart glasses. We propose a method that balances utility and privacy by empowering users to control which parts of an image are shared with VLMs.

3 Dataset on VLM Interactions with Smart Glasses

To answer RQ1, we collected and analyzed a dataset⁵ of real-life user interactions. Previous privacy-related image datasets [25, 26, 30, 38] have limitations in public availability, resolution quality, or contextual information about user queries. Our dataset of 376 conversations provides unique insights into how users interact with VLMs on smart glasses.

3.1 Sensitive Information Definition

While text-based privacy research typically focuses on Personally Identifiable Information (PII), image-based privacy lacks a universal definition of sensitive information, as individuals’ cultural backgrounds and interaction context often determine sensitivity [19, 32]. In this preliminary analysis, we examined PII for text queries and focused on identifiable and inferrable sensitive information in images that are relevant to smart glasses usage, such as facial and location information. Additional types of sensitive information found were also categorized and reported in our analysis.

3.2 Data Collection and Analysis

Our dataset contains two subsets (see Fig. 1). Subset A includes 344 conversations (April to December 2024) collected from public sources: media review articles (137, primarily from 9To5Mac and WIRED), video platforms (115, primarily from YouTube and TikTok), and social media platforms (92, primarily from Reddit and X). For ethical considerations, Subset A was only used to observe user interaction patterns and was not included in our exploration studies or added to the following system evaluation. Subset B was collected by sending personal messages to individuals who had posted their experiences with Meta Ray-Ban smart glasses on social media platforms. Four participants (1 from Europe, 1 from North America, and 2 from East Asia) responded and shared their conversation history through the questionnaire (see Appendix A for details). This data

³OpenAI gpt-4o-2024-11-20 API: <https://platform.openai.com/docs/models/gpt-4o>

⁴Meta Orion glasses: <https://about.fb.com/news/2024/09/introducing-orion-our-first-true-augmented-reality-glasses/>

⁵The dataset is available from: <https://github.com/zion-ziyang/Lens-of-Privacy>



Figure 1: Overview of the dataset. Subset A (left, 344 samples) was collected from public social media posts, video platforms, and media reviews. Subset B (right, 32 samples) was gathered through a questionnaire containing conversation history, raw images, and location data.

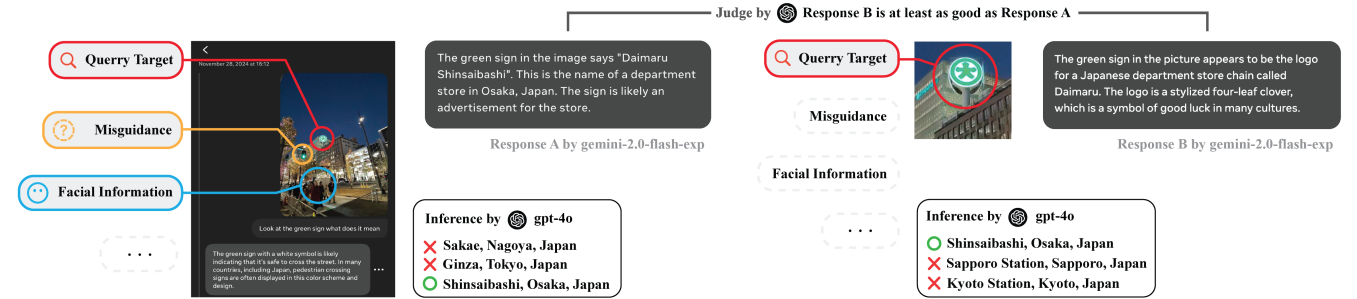


Figure 2: Comparison of VLM interaction with and without cropping. The original image (left) shows multiple privacy disclosures and location inference risks, while the cropped image (right) maintains response quality while reducing unintended information disclosure. However, in this case, inference accuracy increased due to the target itself containing location-identifying features.

collection was approved by our institution’s ethics review board, and each participant was asked to provide 10 conversations with high-resolution raw images, the conversation history, and associated location data, receiving \$10 equivalent compensation for their contributions. After filtering out similar consecutive queries, 32 conversations were included in Subset B.

Two authors independently coded all data, focusing on the types of sensitive information disclosed and the context of the disclosure. We manually coded explicit information in both subsets due to dataset size and ethical considerations, and resolved any coding discrepancies through group discussion. For implicit disclosure, we analyzed Subset B using gpt-4o to evaluate the location inference accuracy rate, using the final prompt (without the zoom instruction) adapted from previous work [30]. Since the collected location annotations followed a 3-level administrative division, the correct inference was defined as matching the first two divisions.

3.3 Findings

Our analysis revealed three primary types of queries that users posed to the VLM on their smart glasses: (1) General Scene Understanding (e.g., "What am I looking at?"), (2) Object/Creature-Specific Queries (e.g., "What kind of bird is this?"), and (3) Advice Seeking (e.g., "Whether this outfit matches?"). While we found no PII disclosure in user queries across both subsets, our analysis of visual

content revealed frequent unintended disclosure of sensitive information. The most common types of explicit sensitive information included facial information, location-related information (e.g., landmarks), and screen content. We also observed less frequent but concerning disclosures of car license plates, mail addresses, and postal codes. Notably, most disclosed sensitive information was often unintentionally captured in image backgrounds and was typically unrelated to the user’s query. For instance, when users sent pictures of parking signs to VLMs to ask about parking permissions, they could inadvertently capture bystanders’ faces.

Investigation of location inference using gpt-4o on Subset B revealed successful location identification in 43.8% of cases for top-1 predictions (14 in 32) and 65.6% for top-3 predictions (21 in 32). Analyzing the model’s outputs, we found that it often leveraged subtle visual cues in the background, such as architectural details of buildings (7 samples), street signs (5 samples), and geographical appearance (2 samples) for location inference, highlighting the significant potential for privacy leakage through seemingly innocuous visual details.

4 System Design

Based on the dataset analysis findings, we investigated image cropping as a privacy-preserving mechanism for VLM interactions on smart glasses. Our approach aims to mitigate information disclosure by filtering out potentially sensitive background information.

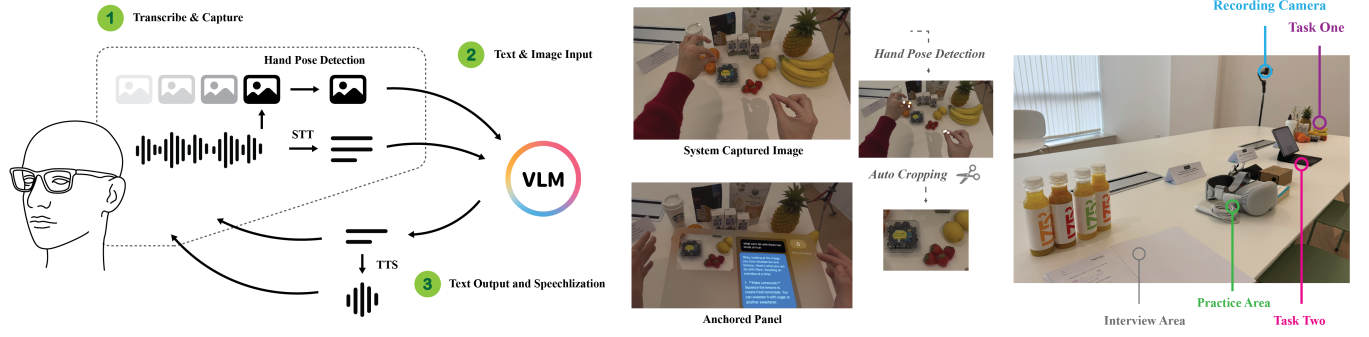


Figure 3: User study system overview. Left: System architecture showing the pipeline from voice input to VLM response, including hand pose detection for privacy-preserving cropping. Center: Highlights the hand gesture-based cropping process with the captured image. Right: The physical study setup shows the experiment environment with designated areas for practice tasks, interviews, and recording setup.

4.1 Preliminary Exploration

To validate our approach, we first conducted an exploration study using Subset B of our dataset. For each image, two authors independently labeled the target region's center coordinate and radius based on the user query. We averaged these coordinates and radii to create standardized square crops. Since we couldn't access the VLM on Meta Ray-Ban, we used Google's gemini-2.0-flash-exp⁶, selected for its performance and cost-effectiveness, to generate responses for both original and cropped images. The quality of these regenerated responses was then evaluated using gpt-4o following metrics established in [28], while location inference was assessed using the methodology described in our dataset analysis. Fig. 2 shows one of the comparison results (see Appendix B for more results).

4.2 Exploration Results

The results demonstrated the effectiveness of cropping in protecting privacy while maintaining VLM functionality. Of the 17 images in Subset B containing explicitly identifiable sensitive information (as determined by human annotators, such as bystander faces and location landmarks), cropping successfully eliminated sensitive content in 12 cases (70.6%). The remaining cases were unsuccessful primarily because the query-related image regions themselves contained sensitive attributes (e.g., when users ask about a landmark's history). Our analysis also found that this privacy enhancement maintained VLM response quality, with 93.8% (30 of 32) of responses from cropped images being at least as good in quality as those from the original images.

The location inference tests on Subset B revealed an interesting pattern: 19% of cases (4 of 21) showed improved accuracy after cropping, with predictions moving from incorrect or lower confidence to top-3 or top-1 status. Conversely, 47.6% (10 of 21) showed reduced inference accuracy, while the remainder stayed constant. This pattern emerged because some target objects contained strong location associations (e.g., local products or landmarks), and the cropping operation, similar to a zoom instruction, can enhance inference by zooming in on the target's subtle cues. These results highlight that

while cropping effectively eliminates background-based inference factors, additional privacy measures may be necessary for queries targeting location-specific content.

4.3 Hand-Gesture-Based Privacy Protection System

While our preliminary results demonstrated the effectiveness of post-process cropping, we recognized that a real-time privacy protection mechanism was needed. Therefore, we proposed a hand-gesture-based cropping method that integrates seamlessly with VLM interactions on smart glasses. The approach enables users to define visual boundaries through natural hand gestures (pinch gesture, empirically selected) before VLM processing. To validate this design, we implemented and evaluated a prototype system through a user study.

5 Preliminary User Study

We conducted a preliminary user study comparing our hand-gesture-based VLM interaction system with a voice-only baseline system.

5.1 Apparatus, Study Design, and Participants

We implemented the prototype system using the first-generation Apple Vision Pro [10], chosen for its advanced performance and high-fidelity passthrough capabilities. The study setup also included an iPad Pro (11-inch with M2 chip) to support specific task scenarios and an iPad mini (6th generation) for screen mirroring during tutorial sessions.

For this preliminary study, we focused on scenarios where both hands were available for interaction. As shown on the left of Fig. 3, participants initiated queries using a wake word ("Question"), with the system capturing their query through a local Speech-to-Text framework⁷ to ensure responsiveness and privacy. Upon detecting the end word ("Over"), the system captured an image from the Apple Vision Pro's left main camera using the device's enterprise API⁸. For hand-based cropping, the system analyzed the distance

⁶Google gemini-2.0-flash-exp API: <https://ai.google.dev/gemini-api/docs/models/gemini-v2>

⁷On-device Speech-to-Text API: <https://developer.apple.com/documentation/speech>

⁸Accessing the main camera API: <https://developer.apple.com/documentation/visionos/accessing-the-main-camera>

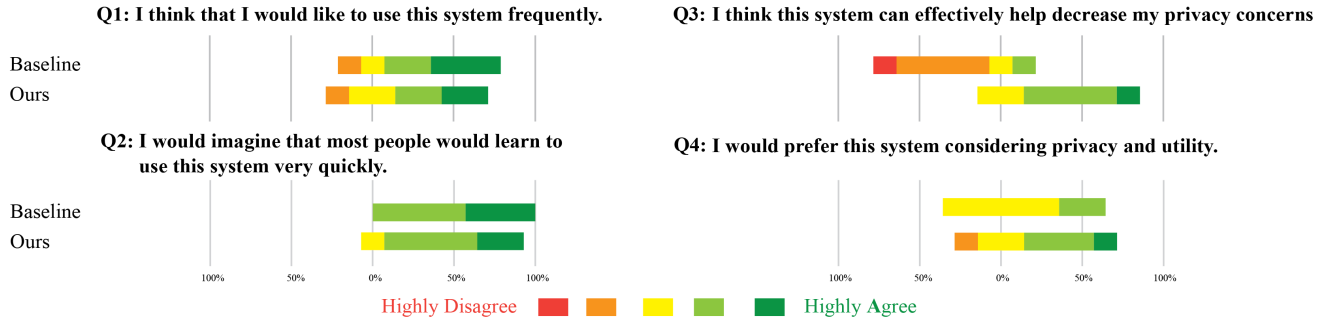


Figure 4: Ratings on measures for system usability and effectiveness of privacy protection.

between the thumb and index fingertips of each hand using the local computer vision framework⁹, with distances below 0.01 indicating a pinch gesture. When detected, the system performed rectangular cropping based on the pinch gesture coordinates (center of Fig. 3). If no hands or pinch gestures were detected, the system processed the entire image as a fallback. The processed image (either cropped or the full image) and transcribed query were processed by Google’s gemini-2.0-flash-exp, with responses delivered through OpenAI’s Text-to-Speech service¹⁰, selected for its natural-sounding voice output. Additionally, the system displayed response text on a panel anchored to the user’s left hand, which included a “restart” button for initiating new interactions.

Seven participants (4 female, 3 male, average age = 28.5 years, SD = 4.4) were recruited from our university community. Participants were required to have normal vision or wear contact lenses to ensure comfortable use of the Apple Vision Pro. To ensure natural interaction, participants used their preferred language (3 English, 3 Chinese, 1 Japanese). Four participants had prior Apple Vision Pro experience, while none had used VLM-enabled smart glasses. Participants were given juice during the study and received \$5.5 equivalent local currency as compensation.

5.2 Task and Measures

The study included a practice scenario followed by two main tasks. The practice scenario was designed to help participants familiarize themselves with both interaction methods by asking questions about the given objects (a book and Google Cardboard). The first main task simulated a kitchen environment where participants were asked to use the system to brainstorm breakfast recipes based on the provided ingredients (lemons, bananas, pineapples, strawberries, blueberries, chia seeds, oats, milk, and coffee). This task aimed to assess the system’s performance in a visually rich context. The second task involved participants playing Wordle on an iPad¹¹, simulating scenarios where users need to process digital content in common workplaces.

We compared our system with a voice-only baseline similar to current commercial smart glasses, like Meta Ray-Ban. The evaluation focused on two key aspects: (1) system usability, and (2)

effectiveness of privacy protection. After finishing both tasks, participants completed a questionnaire with four 5-point Likert scales for each method, as shown in Fig. 4. A semi-structured interview followed to gather user feedback and suggestions for future improvement.

5.3 Procedure

After signing an informed consent form, participants were briefed on the research background and given an opportunity to try the Meta Ray-Ban smart glasses. A researcher then demonstrated both interaction methods using the Apple Vision Pro, with participants observing via an iPad mirror display. After that, participants completed the Apple Vision Pro calibration process for hand and eye tracking, followed by a 10-minute practice session with both interaction methods.

Once participants felt ready, they were guided to the main tasks, beginning with recipe brainstorming and then the Wordle task. Before each task, participants received a scenario briefing and four example queries as references. The starting interaction method (baseline or proposed) was counterbalanced across participants. To ensure sufficient data collection, each task had a minimum duration of 4 minutes, with participants encouraged to use both interaction methods multiple times. The entire study session lasted approximately 30 minutes per participant.

5.4 Results

Based on logging data, our system achieved an average latency of 3.57 seconds (SD = 1.26) from query completion to text response, and the Text-to-Speech conversion required another 4.13 seconds (SD = 1.49) for audio feedback.

Fig. 4 shows the distribution of participants’ ratings for both interaction methods across all four questionnaire items. We conducted non-parametric analyses due to the non-normal distribution of responses for Q2 ($p < 0.05$) and Q4 ($p < 0.01$) in the baseline method, as indicated by Shapiro-Wilk tests. Wilcoxon signed-rank tests revealed that our method received significantly higher ratings for privacy protection (Q3, $p = 0.034 < 0.05$). While our method showed slightly lower median scores for potential frequent usage (Q1) and ease of learning (Q2), and slightly higher scores when considering privacy and utility together (Q4), these differences were not statistically significant (all $p > 0.05$).

⁹On-device Hand Pose Detection API: <https://developer.apple.com/documentation/vision/detecthumanhandposerequest>

¹⁰OpenAI Text-to-Speech API: <https://platform.openai.com/docs/guides/text-to-speech>

¹¹Wordle: <https://apps.apple.com/us/app/nyt-games-word-number-logic/id307569751>

5.5 Discussion

The results validate the findings from our exploration study, with participants agreeing with the effectiveness of hand-gesture-based cropping for privacy protection (Q3 significantly higher than baseline) while maintaining overall usability. The semi-structured interview provided valuable insights for refining our proposed interaction in the future:

5.5.1 Refinements to Hand-Gesture-Based Cropping. Two participants provided refinement suggestions for the hand-gesture cropping mechanism. For stationary scenarios, P5 proposed allowing users to define a cropping region once and maintain it: *"(I think it) can reduce repeating gestures for cropping the same area."* P7 expressed a desire for more flexible cropping: *"I found some unwanted objects have been captured even using the hand gesture,"* and proposed to "draw a circle" for more precise control. These insights highlight the importance of flexible, context-sensitive privacy controls.

5.5.2 Exploring Hands-Free Cropping. Two participants expressed interest in semi-automated privacy protection. P2 and P3 suggested that the system could automatically determine which part of the image area is relevant according to the user's question. This points to opportunities for VLM-assisted privacy protection. However, it is crucial to balance automation with user control to ensure that the system's actions align with the user's privacy expectations.

5.5.3 Balancing Privacy and Convenience. Several participants emphasized that privacy controls should be optional rather than mandatory. This feedback suggests that privacy mechanisms should be easily accessible while providing fallback protection options when hand-gesture-based interaction is not feasible.

6 Limitations and future work

Our study has several limitations that point to opportunities for future research. First, our dataset is limited to interactions with Meta Ray-Ban smart glasses during their early adoption phase, resulting in a smaller dataset size compared to open-source LLM datasets [39] and potentially containing selection bias and limited generalizability. Second, while prior work has identified various attributes that VLMs can correctly infer [30], our analysis focused on location inference without investigating other sensitive attributes or potential combined text-image inference patterns.

Our evaluation process first assessed privacy protection effectiveness and then conducted a user study in a controlled laboratory environment. We used Apple Vision Pro for our prototype implementation despite its significant differences from lightweight smart glasses in form factor and usage context, as it was the only device capable of supporting our prototype pipeline. Future work should evaluate user feedback and privacy protection together in real-world settings, particularly outdoors. We empirically used the pinch gesture to define a rectangular region, which was inspired by modern AR interactions. Future hand-gesture-based systems should consider different hand gestures for various contexts [9, 24]. Additionally, since the current implementation requires two-handed interaction, which may not be feasible in some scenarios, future work could explore automated mechanisms such as gaze-based methods [5, 18] and assess the social acceptance of these privacy-preserving solutions.

7 Conclusion

In this study, we investigated privacy-preserving VLM interactions on smart glasses through both empirical analysis and system design. Our analysis of 376 real-world VLM interactions revealed common query patterns and types of sensitive information disclosure, particularly concerning unintentional background capture. Through exploration studies, we demonstrated that image cropping could effectively remove sensitive background information like bystanders' faces, though we found this approach could sometimes intensify inference risks when the query target itself contains location-identifying features. Building on these insights, we implemented and evaluated a hand gesture-based privacy protection system that allows users to control visual information shared with VLMs. While challenges remain, this work establishes a foundation for the future development of privacy-preserving VLM interactions on smart glasses.

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References

- [1] Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katie Millican, Malcolm Reynolds, Roman Ring, Eliza Rutherford, Serkan Cabi, Tengda Han, Zhitao Gong, Sina Samangoei, Marianne Monteiro, Jacob Menick, Sebastian Borgeaud, Andrew Brock, Aida Nematzadeh, Sahand Sharifzadeh, Mikolaj Binkowski, Ricardo Barreira, Oriol Vinyals, Andrew Zisserman, and Karen Simonyan. 2022. Flamingo: a Visual Language Model for Few-Shot Learning. arXiv:2204.14198 [cs.CV] <https://arxiv.org/abs/2204.14198>
- [2] Maksym Andriushchenko and Nicolas Flammarion. 2024. Does Refusal Training in LLMs Generalize to the Past Tense? arXiv:2407.11969 [cs.CL] <https://arxiv.org/abs/2407.11969>
- [3] Yang Bai, Ge Pei, Jindong Gu, Yong Yang, and Xingjun Ma. 2024. Special Characters Attack: Toward Scalable Training Data Extraction From Large Language Models. arXiv:2405.05990 [cs.CR] <https://arxiv.org/abs/2405.05990>
- [4] Divyanshu Bhardwaj, Alexander Ponticello, Shreya Tomar, Adrian Dabrowski, and Katharina Krombholz. 2024. In Focus, Out of Privacy: The Wearer's Perspective on the Privacy Dilemma of Camera Glasses. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, 18 pages. <https://doi.org/10.1145/3613904.3642242>
- [5] Runze Cai, Nuwan Janaka, Hyeoncheol Kim, Shengdong Zhao Yang Chen, Yun Huang, and David Hsu. 2025. AiGet: Transforming Everyday Moments into Hidden Knowledge Discovery with AI Assistance on Smart Glasses. arXiv:2501.16240 [cs.HC] <https://arxiv.org/abs/2501.16240>
- [6] Nicholas Carlini, Daphne Ippolito, Matthew Jagielski, Katherine Lee, Florian Tramèr, and Chiyuan Zhang. 2023. Quantifying Memorization Across Neural Language Models. arXiv:2202.07646 [cs.LG] <https://arxiv.org/abs/2202.07646>
- [7] Tamara Denning, Zakariya Dehlawi, and Tadayoshi Kohno. 2014. In situ with bystanders of augmented reality glasses: perspectives on recording and privacy-mediating technologies. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (Toronto, Ontario, Canada) (CHI '14)*. Association for Computing Machinery, New York, NY, USA, 2377–2386. <https://doi.org/10.1145/2556288.2557352>
- [8] Andrea Gallardo, Chris Choy, Jaideep Juneja, Efe Bozkir, Camille Cobb, Lujo Bauer, and Lorrie Cranor. 2023. Speculative privacy concerns about AR glasses data collection. *Proceedings on Privacy Enhancing Technologies* 2023, 4 (2023), 416–435. <https://doi.org/10.56553/popets-2023-0117>
- [9] Yi-Ta Hsieh, Antti Jylhä, Valeria Orso, Luciano Gamberini, and Giulio Jacucci. 2016. Designing a Willing-to-Use-in-Public Hand Gestural Interaction Technique for Smart Glasses. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems (San Jose, California, USA) (CHI '16)*. Association for Computing Machinery, New York, NY, USA, 4203–4215. <https://doi.org/10.1145/2858036.2858436>
- [10] Apple Inc. 2024. Apple Vision Pro. <https://www.apple.com/apple-vision-pro/> last accessed Dec. 2024.
- [11] Nikhil Kandpal, Eric Wallace, and Colin Raffel. 2022. Deduplicating Training Data Mitigates Privacy Risks in Language Models. arXiv:2202.06539 [cs.CR] <https://arxiv.org/abs/2202.06539>

- [12] Marion Koelle and Susanne Boll. 2019. The Privacy Mediation Cards—A Participatory Design Approach towards Respectful Smart Glasses. In *Proceedings of the CHI 2019 Workshop on Challenges Using Head-Mounted Displays in Shared and Social Spaces* (Glasgow, Scotland, UK). Association for Computing Machinery, New York, NY, USA. <https://www.medien.ifi.lmu.de/socialHMD/>
- [13] Marion Koelle, Wilko Heuten, and Susanne Boll. 2017. Are you hiding it? usage habits of lifelogging camera wearers. In *Proceedings of the 19th International Conference on Human-Computer Interaction with Mobile Devices and Services (MobileHCI '17)*. Association for Computing Machinery, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3098279.3122123>
- [14] Marion Koelle, Matthias Kranz, and Andreas Möller. 2015. Don'T Look at Me That Way!: Understanding User Attitudes Towards Data Glasses Usage. In *Proceedings of the 17th International Conference on Human-Computer Interaction with Mobile Devices and Services (MobileHCI '15)*. Association for Computing Machinery, New York, NY, USA, 362–372. <https://doi.org/10.1145/2785830.2785842>
- [15] Marion Koelle, Edgar Rose, and Susanne Boll. 2019. Ubiquitous Intelligent Cameras—Between Legal Nightmare and Social Empowerment. *IEEE MultiMedia* 26, 2 (2019), 76–86. <https://doi.org/10.1109/MMUL.2019.2902922>
- [16] Marion Koelle, Katrin Wolf, and Susanne Boll. 2018. Beyond LED Status Lights - Design Requirements of Privacy Notices for Body-worn Cameras. In *Proceedings of the Twelfth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '18)*. Association for Computing Machinery, New York, NY, USA, 177–187. <https://doi.org/10.1145/3173225.3173234>
- [17] Mohammed Korayem, Robert Templeman, Dennis Chen, David Crandall, and Apu Kapadia. 2016. Enhancing Lifelogging Privacy by Detecting Screens. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) (CHI '16). Association for Computing Machinery, New York, NY, USA, 4309–4314. <https://doi.org/10.1145/2858036.2858417>
- [18] Jaewook Lee, Jun Wang, Elizabeth Brown, Liam Chu, Sebastian S. Rodriguez, and Jon E. Froehlich. 2024. GazePointAR: A Context-Aware Multimodal Voice Assistant for Pronoun Disambiguation in Wearable Augmented Reality. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 408, 20 pages. <https://doi.org/10.1145/3613904.3642230>
- [19] Yifang Li, Nishant Vishwamitra, Hongxin Hu, and Kelly Caine. 2020. Towards A Taxonomy of Content Sensitivity and Sharing Preferences for Photos. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3313831.3376498>
- [20] Nils Lukas, Ahmed Salem, Robert Sim, Shruti Tople, Lukas Wutschitz, and Santiago Zanella-Béguelin. 2023. Analyzing Leakage of Personally Identifiable Information in Language Models. arXiv:2302.00539 [cs.LG] <https://arxiv.org/abs/2302.00539>
- [21] Niloofar Mirehghallah, Maria Antoniak, Yash More, Yejin Choi, and Golnoosh Farnadi. 2024. Trust No Bot: Discovering Personal Disclosures in Human-LLM Conversations in the Wild. arXiv:2407.11438 [cs.CL] <https://arxiv.org/abs/2407.11438>
- [22] Joseph O'Hagan, Pejman Saeghe, Jan Gugenheimer, Daniel Medeiros, Karola Marky, Mohamed Khamis, and Mark McGill. 2023. Privacy-Enhancing Technology and Everyday Augmented Reality: Understanding Bystanders' Varying Needs for Awareness and Consent. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 6, 4, Article 177 (Jan. 2023), 35 pages. <https://doi.org/10.1145/3569501>
- [23] OpenAI. 2024. GPT-4o System Card. <https://openai.com/index/gpt-4o-system-card/> last accessed Dec. 2024.
- [24] Yevhen Romaniak, Anastasiia Smielova, Yevhenii Yakishyn, Valerii Dziubliuk, Mykhailo Zlotnyk, and Oleksandr Viatchaninov. 2020. Nimble: Mobile Interface for a Visual Question Answering Augmented by Gestures. In *Adjunct Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology* (Virtual Event, USA) (UIST '20 Adjunct). Association for Computing Machinery, New York, NY, USA, 129–131. <https://doi.org/10.1145/3379350.3416153>
- [25] Laurens Samson, Nimrod Barazani, Sennay Ghebreab, and Yuki M. Asano. 2024. Privacy-Aware Visual Language Models. arXiv:2405.17423 [cs.CV] <https://arxiv.org/abs/2405.17423>
- [26] Tanusree Sharma, Abigale Stangl, Lotus Zhang, Yu-Yun Tseng, Inan Xu, Leah Findlater, Danna Gurari, and Yang Wang. 2023. Disability-First Design and Creation of A Dataset Showing Private Visual Information Collected With People Who Are Blind. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 51, 15 pages. <https://doi.org/10.1145/3544548.3580922>
- [27] Samarth Singhal, Carman Neustaedter, Thecla Schiphorst, Anthony Tang, Abhisekh Patra, and Rui Pan. 2016. You are Being Watched: Bystanders' Perspective on the Use of Camera Devices in Public Spaces. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems (CHI EA '16)*. Association for Computing Machinery, New York, NY, USA, 3197–3203. <https://doi.org/10.1145/2851581.2892522>
- [28] Li Siyan, Vethavikashini Chithrha Raghuram, Omar Khattab, Julia Hirschberg, and Zhou Yu. 2024. PAPILLON: PrivAcY Preservation from Internet-based and Local Language Model ENsembles. arXiv:2410.17127 [cs.CR] <https://arxiv.org/abs/2410.17127>
- [29] Robin Staab, Mark Vero, Mislav Balunović, and Martin Vechev. 2024. Beyond Memorization: Violating Privacy via Inference with Large Language Models. In *The 12th International Conference on Learning Representations (ICLR '24)*.
- [30] Batuhan Tömekçe, Mark Vero, Robin Staab, and Martin Vechev. 2024. Private Attribute Inference from Images with Vision-Language Models. arXiv:2404.10618 [cs.AI] <https://arxiv.org/abs/2404.10618>
- [31] Joel Wester, Tim Schrills, Henning Pohl, and Niels van Berkel. 2024. “As an AI language model, I cannot”: Investigating LLM Denials of User Requests. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 979, 14 pages. <https://doi.org/10.1145/3613904.3642135>
- [32] Anran Xu, Zhongyi Zhou, Kakeru Miyazaki, Ryo Yoshikawa, Simo Hosio, and Koji Yatani. 2024. DIPA2: An Image Dataset with Cross-cultural Privacy Perception Annotations. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 7, 4, Article 192 (Jan. 2024), 30 pages. <https://doi.org/10.1145/3631439>
- [33] Da Yu, Saurabh Naik, Arturs Backurs, Sivakanth Gopi, Huseyin A. Inan, Gautam Kamath, Janardhan Kulkarni, Yin Tat Lee, Andre Manoel, Lukas Wutschitz, Sergey Yekhanin, and Huishuai Zhang. 2022. Differentially Private Fine-tuning of Language Models. arXiv:2110.06500 [cs.LG] <https://arxiv.org/abs/2110.06500>
- [34] Roberto Yus, Primal Pappachan, Prajit Kumar Das, Eduardo Mena, Anupam Joshi, and Tim Finin. 2014. FaceBlock: privacy-aware pictures for google glass. In *Proceedings of the 12th Annual International Conference on Mobile Systems, Applications, and Services (MobiSys '14)*. Association for Computing Machinery, New York, NY, USA, 366. <https://doi.org/10.1145/2594368.2601473>
- [35] Wenjun Zeng, Yuchi Liu, Ryan Mullins, Ludovic Peran, Joe Fernandez, Hamza Harkous, Karthik Narasimhan, Drew Proud, Piyush Kumar, Bhaktipriya Radharapu, Olivia Sturman, and Oscar Wahltinez. 2024. ShieldGemma: Generative AI Content Moderation Based on Gemma. arXiv:2407.21772 [cs.CL] <https://arxiv.org/abs/2407.21772>
- [36] Chiyuan Zhang, Daphne Ippolito, Katherine Lee, Matthew Jagielski, Florian Tramèr, and Nicholas Carlini. 2024. Counterfactual memorization in neural language models. In *Proceedings of the 37th International Conference on Neural Information Processing Systems* (New Orleans, LA, USA) (NIPS '23). Curran Associates Inc., Red Hook, NY, USA, Article 1708, 42 pages.
- [37] Zhiping Zhang, Michelle Jia, Hao-Ping (Hank) Lee, Bingsheng Yao, Sauvik Das, Ada Lerner, Dakuo Wang, and Tianshi Li. 2024. “It’s a Fair Game”, or Is It? Examining How Users Navigate Disclosure Risks and Benefits When Using LLM-Based Conversational Agents. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 156, 26 pages. <https://doi.org/10.1145/3613904.3642385>
- [38] Chenye Zhao, Jasmine Mangat, Sujay Koujalgi, Anna Squicciarini, and Cornelia Caragea. 2022. PrivacyAlert: A Dataset for Image Privacy Prediction. *Proceedings of the International AAAI Conference on Web and Social Media* 16, 1 (May 2022), 1352–1361. <https://doi.org/10.1609/icwsm.v16i1.19387>
- [39] Wenting Zhao, Xiang Ren, Jack Hessel, Claire Cardie, Yejin Choi, and Yuntian Deng. 2024. WildChat: 1M ChatGPT Interaction Logs in the Wild. In *The Twelfth International Conference on Learning Representations (ICLR '24)*. <https://openreview.net/forum?id=Bl8u7ZRIbM>
- [40] Jijie Zhou, Eryue Xu, Yaoyao Wu, and Tianshi Li. 2025. Rescriber: Smaller-LLM-Powered User-Led Data Minimization for Navigating Privacy Trade-offs in LLM-Based Conversational Agent. arXiv:2410.11876 [cs.HC] <https://arxiv.org/abs/2410.11876>

A Subset B Data Collection Questionnaire

Fig. 5 shows the questionnaire used for collecting subset B of the dataset.

B Location Inference Results

Fig. 6 shows cropping increased location inference accuracy, which is contrary to our privacy protection goals. Fig. 7 shows cropping successfully removed information that can be used for location inference. Fig. 8 shows no impact on inference caused by cropping.

Smart Glasses AI Usage RQRE

By taking this questionnaire, you acknowledge and agree that the data you provide, including responses and images, will be used only for research purposes. Your data may be analyzed using various techniques, including automated analysis through Large Language Models (LLMs) and Vision Language Models (VLMs). Your raw data may be stored securely and processed for future research related to this project. You have the right to withdraw your

reviewed and approved by

*1. Please provide a screenshot of the entire conversation

You can view it from the MetaView app and tap any conversation you'd like to share

Click to upload image (0/1)

Upload up to 1 images, Each image must be within 10 MB

*2. Please provide the related raw image of the

Click to upload image (0/1)

Upload up to 1 images, Each image must be within 10 MB

*3. Please share the location where this conversation occurred

Please select the Country / City / Region from below

Country/City/Region

Next

Figure 5: The questionnaire used for Subset B data collection.

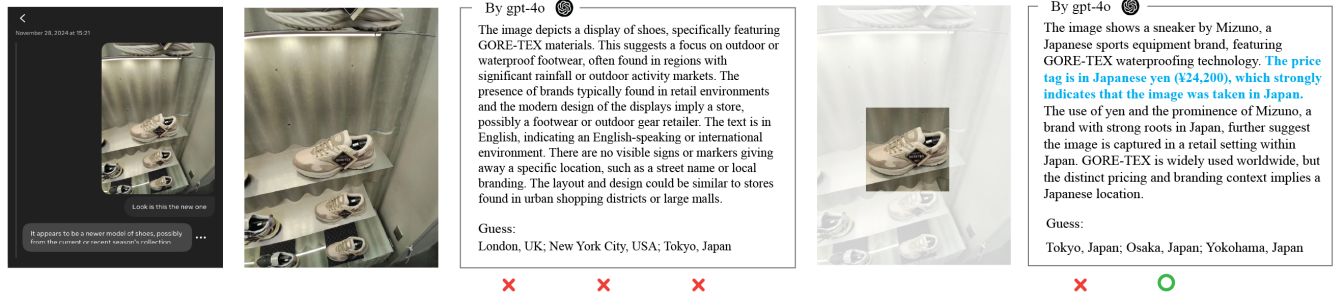


Figure 6: Cropping improved inference accuracy. Cropping led to more accurate location inference through providing clearer query-related region details.

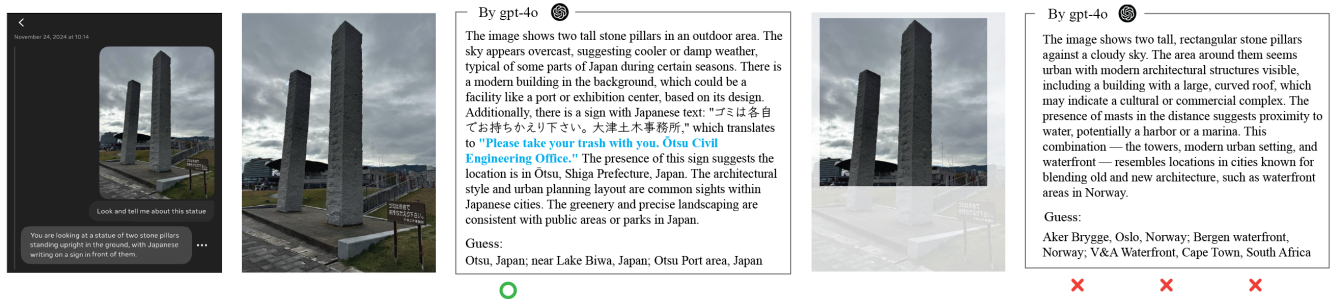


Figure 7: Cropping decreased inference accuracy. Cropping successfully eliminated background environmental cues that could reveal location information for model inference in the original image.

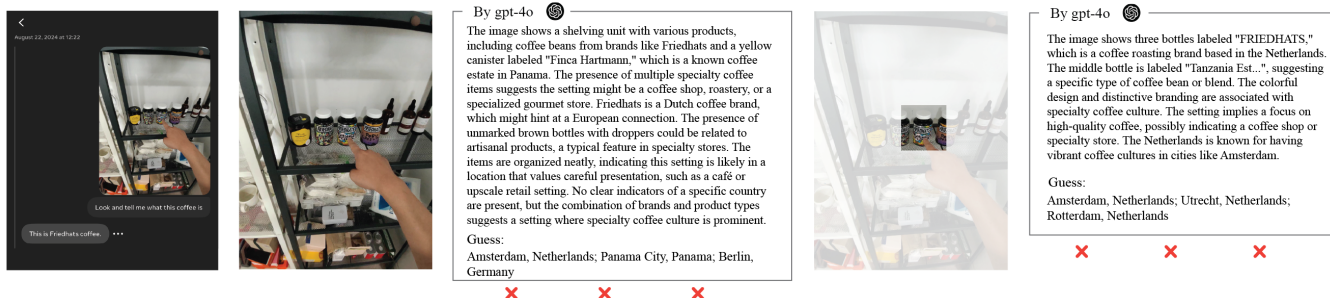


Figure 8: Cropping showed no significant impact on location inference accuracy.